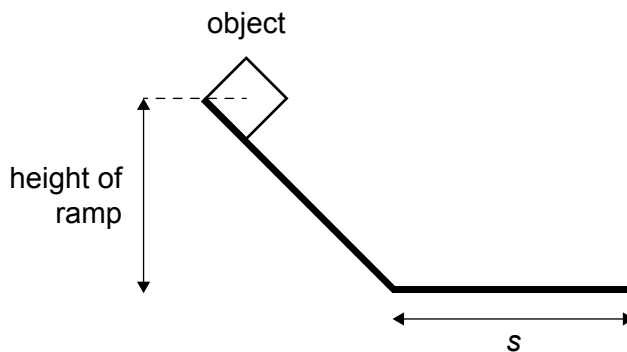


PAPER 1 / QUESTION 117

25 April 2024 / TZ1 (file label) / HL Paper 1 / original question 3 / 2224-6701

- 117.** An object is released from rest and slides down a frictionless ramp. The object then leaves the ramp and slides along a rough horizontal surface. The object stops in a distance s along the ramp.



The coefficient of dynamic friction between the object and the rough horizontal surface is μ .

What is the height of the ramp?

- A. $\mu g s$
- B. $\frac{s}{2g\mu}$
- C. $\frac{s}{\mu}$
- D. μs